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Executive Summary

This report describes the design and implementation of a database for a Gym Management System for a mid size fitness facility. The system is used to manage memberships, trainers, classes, equipment, and payments. It keeps track of members attendance, class bookings, staff schedules, and maintenance on equipment, and payments, etc. The report discusses the case study, ER model, relational schema, business rules, normalisation to 3NF, data dictionary, implementation of the SQL (DDL and DML), and sample screenshots of the implemented database with XAMPP.

Introduction & Assumptions

The Gym Management System database was designed to organize and facilitate business for a fitness centre. As part of the system members are tracked, along with memberships, classes, staff (both trainers and receptionists), equipment maintenance, bookings, and payments. The system was designed based on the following assumptions:

- Each member can hold multiple memberships, however, a member may only have one active membership at any given time.
- Staff may be designated as either Trainers or Receptionists, but are actually a subtype of a Staff supertype.
- Each class has a Trainer scheduled and can have multiple member bookings.
- Equipment maintenance can be performed by any staff member.

Case Study Description

This case study presents a comprehensive overview of a Gym Management System that is designed for usage within a mid sized fitness center. The database manages all critical aspects related to memberships, trainers, classes, equipment, scheduling, and payment processing. The system accurately tracks member attendance, member class bookings, employee schedules, and manages class attendance to ensure effective operation every hour of every day. The system also monitors equipment for maintenance purposes, so that equipment will last longer and be safer for individuals using it. It will also help facilitate an easy payment transaction for gym

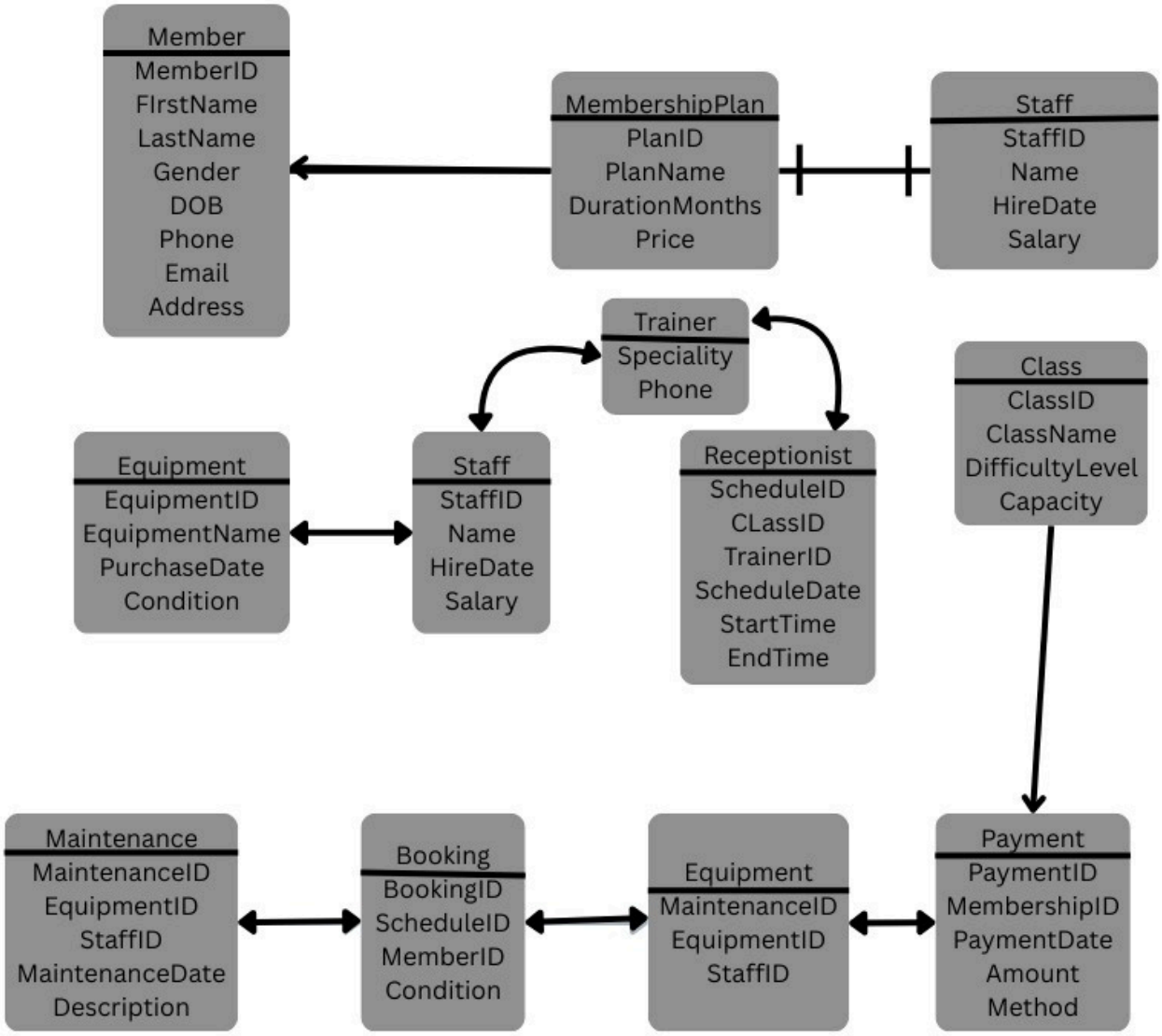
management and members. The Gym Management System is designed to improve operations and increase member participation, which ultimately leads to the profitability and growth of fitness center management operations.

Entities, Attributes & Keys

The following entities, attributes and keys are given bellow:

- Member(MemberID [PK], FirstName, LastName, Gender, DOB, Phone, Email, Address)
- MembershipPlan(PlanID [PK], PlanName, DurationMonths, Price)
- Membership(MembershipID [PK], MemberID [FK], PlanID [FK], StartDate, EndDate, Status)
- Staff(StaffID [PK], Name, HireDate, Salary)
- Trainer(StaffID [PK, FK], Speciality, Phone)
- Receptionist(StaffID [PK, FK], Shift)
- Class(ClassID [PK], ClassName, DifficultyLevel, Capacity)
- ClassSchedule(ScheduleID [PK], ClassID [FK], TrainerID [FK], ScheduledDate, StartTime, EndTime)
- Booking(BookingID [PK], ScheduleID [FK], MemberID [FK], BookingDate, Status)
- Equipment(EquipmentID [PK], EquipmentName, PurchaseDate, Condition)
- Maintenance(MaintenanceID [PK], EquipmentID [FK], StaffID [FK], MaintenanceDate, Description)
- Payment(PaymentID [PK], MembershipID [FK], PaymentDate, Amount, Method)

ER Diagram


Member
MemberID
FIrstName
LastName
Gender
DOB
Phone
Email
Address
MembershipPlan
PlanID
PlanName
DurationMonths
Price
Staff
StaffID
Name
HireDate
Salary
Trainer
Speciality
Phone
Class
ClassID
ClassName
DifficultyLevel
Capacity
Equipment
EquipmentID
EquipmentName
PurchaseDate
Condition
Staff
StaffID
Name
HireDate
Salary
Receptionist
ScheduleID
CLassID
TrainerID
ScheduleDate
StartTime
EndTime
Maintenance
MaintenanceID
EquipmentID
StaffID
MaintenanceDate
Description
Booking
BookingID
ScheduleID
MemberID
Condition
Equipment
MaintenanceID
EquipmentID
StaffID
Payment
PaymentID
MembershipID
PaymentDate
Amount
Method

Relationship Schema

The relational schema for the Gym Management System is given bellow :

- Member(MemberID [PK], FirstName, LastName, Gender, DOB, Phone, Email, Address)
- MembershipPlan(PlanID [PK], PlanName, DurationMonths, Price)
- Membership(MembershipID [PK], MemberID [FK], PlanID [FK], StartDate, EndDate, Status)
- Staff(StaffID [PK], Name, HireDate, Salary)
- Trainer(StaffID [PK, FK], Speciality, Phone)
- Receptionist(StaffID [PK, FK], Shift)
- Class(ClassID [PK], ClassName, DifficultyLevel, Capacity)
- ClassSchedule(ScheduleID [PK], ClassID [FK], TrainerID [FK], ScheduledDate, StartTime, EndTime)
- Booking(BookingID [PK], ScheduleID [FK], MemberID [FK], BookingDate, Status)
- Equipment(EquipmentID [PK], EquipmentName, PurchaseDate, Condition)
- Maintenance(MaintenanceID [PK], EquipmentID [FK], StaffID [FK], MaintenanceDate, Description)
- Payment(PaymentID [PK], MembershipID [FK], PaymentDate, Amount, Method)

Business Rules with Cardinalities and Connectivity

The following business rules are given the relationships between entities:

- Member to Membership: One-to-Many (Mandatory).
- Member to ClassSchedule: Many-to-Many resolved through Booking (Mandatory).
- MembershipPlan to Membership: One-to-Many (Mandatory).
- Trainer to ClassSchedule: One-to-Many (Mandatory).
- Class to ClassSchedule: One-to-Many (Mandatory).

- Equipment to Maintenance: One-to-Many (Optional).
- Staff to Maintenance: One-to-Many (Optional).
- Membership to Payment: One-to-Many (Mandatory).

Normalization Steps (3NF) with Dependency Diagrams

- **Member**

  Functional Dependency Diagram for Member:

  MemberID → FirstName, LastName, Gender, DOB, Phone, Email, Address
- **MembershipPlan**

  Functional Dependency Diagram for MembershipPlan:

  PlanID → PlanName, DurationMonths, Price
- **Membership**

  Functional Dependency Diagram for Membership:

  MembershipID → MemberID, PlanID, StartDate, EndDate, Status
- **Staff**

  Functional Dependency Diagram for Staff:

  StaffID → Name, HireDate, Salary
- **Trainer**

  Functional Dependency Diagram for Trainer:

  StaffID → Speciality, Phone
- **Receptionist**

  Functional Dependency Diagram for Receptionist:

  StaffID → Shift
- **Class**

  Functional Dependency Diagram for Class:

  ClassID → ClassName, DifficultyLevel, Capacity
- **ClassSchedule**

  Functional Dependency Diagram for ClassSchedule:

  ScheduleID → ClassID, TrainerID, ScheduledDate, StartTime, EndTime
- **Booking**

  Functional Dependency Diagram for Booking:

BookingID → ScheduleID, MemberID, BookingDate, Status

- **Equipment**

  Functional Dependency Diagram for Equipment:

  EquipmentID → EquipmentName, PurchaseDate, Condition

- **Maintenance**

  Functional Dependency Diagram for Maintenance:

  MaintenanceID → EquipmentID, StaffID, MaintenanceDate, Description

- **Payment**

  Functional Dependency Diagram for Payment:

  PaymentID → MembershipID, PaymentDate, Amount, Method

Data Dictionary

In Data Dictionary there is given the attributes, data types, constraints, and descriptions for each entity in the Gym Management System database.

MEMBER :

Attribute Name	Data Type	Size	Null / Not Key Null	/ Constraint	Description
MemberID	INT	-	NOT NULL	Primary Key (PK)	Unique identifier for each member
FirstName	VARCHAR	50	NOT NULL		Member's first name
LastName	VARCHAR	50	NOT NULL		Member's last name

Gender	VARCHAR	10	NULL		Gender of member Date
DOB	DATE	-	NULL		of birth
Phone	VARCHAR	20	NULL		Member's phone number
Email	VARCHAR	100	NULL		Member's email address
Address	VARCHAR	150	NULL		Member's residential address

MEMBERSHIPPLAN :

Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description
PlanID	INT	-	NOT NULL	Primary Key (PK)	Unique identifier for each membership plan
PlanName	VARCHAR	50	NOT NULL		Name of the plan

DurationMonths	INT	-	NOT NULL		Duration of the plan in months
Price	DECIMAL	10,2	NOT NULL		Price of the plan

MEMBERSHIP :

Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description
MembershipID	INT	-	NOT NULL	Primary Key (PK)	Unique ID for each membership
MemberID	INT	-	NOT NULL	Foreign Key MEMBER	- The member who holds this membership
PlanID	INT	-	NOT NULL	Foreign Key - MEMBERSHIPPLAN	The plan chosen by the member
StartDate	DATE	-	NULL		Membership start date
EndDate	DATE	-	NULL		Membership end date
Status	VARCHAR	20	NULL		Current status

					(Active, Cancelled)

STAFF :

Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description
StaffID	INT	-	NOT NULL	Primary Key (PK)	Unique identifier for each staff member
Name	VARCHAR	100	NOT NULL		Staff member's name
HireDate	DATE	-	NULL		Date of hire
Salary	DECIMAL	10,2	NULL		Salary of staff

TRAINER :

Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description
StaffID	INT	-	NOT NULL	Primary Key/Foreign Key - STAFF	Link to STAFF entity
Speciality	VARCHAR	100	NULL		Trainer's speciality

Phone	VARCHAR	20	NULL		Trainer's phone number

RECEPTIONIST :

Attribute Name	Data Type	Size	Null / Not Key Null	/ Constraint	Description
StaffID	INT	-	NOT NULL	Primary Key/Foreign Key - STAFF	Link to STAFF entity
Shift	VARCHAR	50	NULL		Receptionist's work shift

CLASS :

Attribute Name	Data Type	Size	Null / Not Key Null	/ Constraint	Description
ClassID	INT	-	NOT NULL	Primary Key (PK)	Unique identifier for each class
ClassName	VARCHAR	100	NOT NULL		Name of the class
DifficultyLevel	VARCHAR	20	NULL		Level of difficulty
Capacity	INT	-	NULL		Capacity of the class

CLASSSCHEDULE :

Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description
ScheduleID	INT	-	NOT NULL	Primary Key (PK)	Unique ID for each scheduled class
ClassID	INT	-	NOT NULL	Foreign Key - CLASS	Class being scheduled
TrainerID	INT	-	NOT NULL	Foreign Key - TRAINER	Trainer conducting the class
ScheduledDate	DATE	-	NULL		Date of the class
StartTime	TIME	-	NULL		Start time
EndTime	TIME	-	NULL		End time

BOOKING :

Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description
BookingID	INT	-	NOT NULL	Primary Key (PK)	Unique booking ID
ScheduleID	INT	-	NOT NULL	Foreign Key - CLASSSCHEDULE	Scheduled class booked

MemberID	INT	-	NOT NULL	Foreign Key - MEMBER	Member making the booking
BookingDate	DATE	-	NULL		Date of booking
Status	VARCHAR	20	NULL		Booking status

EQUIPMENT :

Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description
EquipmentID	INT	-	NOT NULL	Primary Key (PK)	Unique ID for each equipment
EquipmentName	VARCHAR	100	NOT NULL		Name of the equipment
PurchaseDate	DATE	-	NULL		Date of purchase
Situation	VARCHAR	50	NULL		Current situation

MAINTENANCE :

Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description

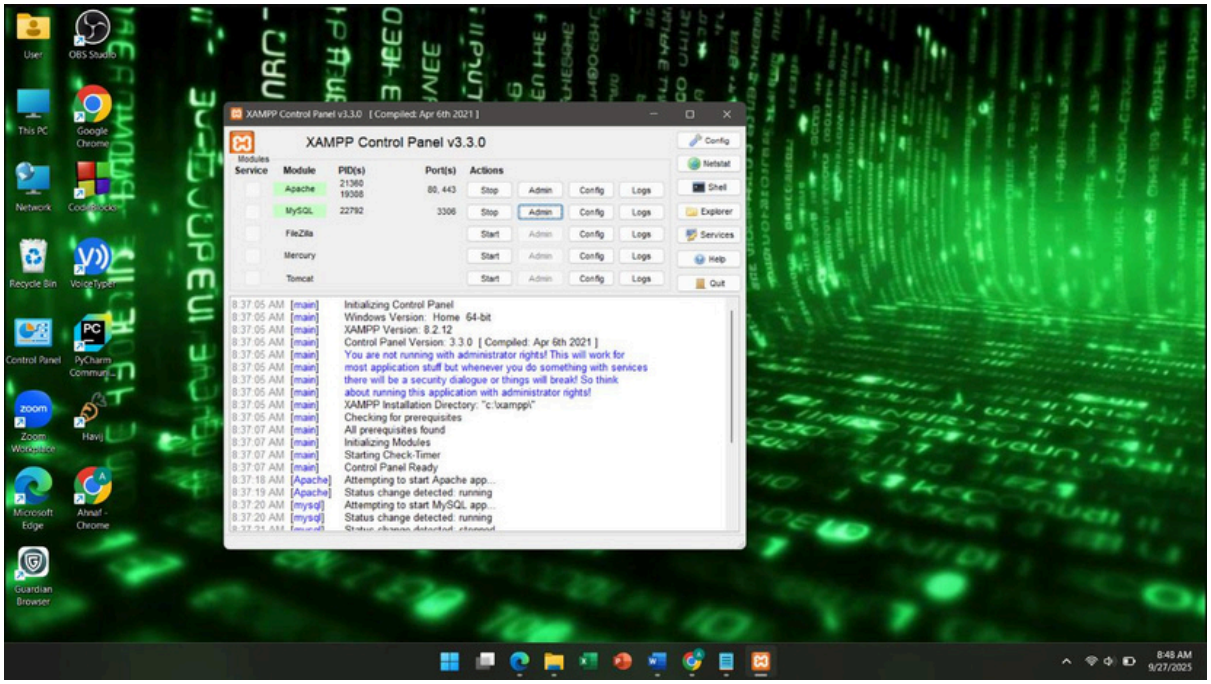
MaintenanceID	INT	-	NOT NULL	Primary Key (PK)	Unique maintenance record
EquipmentID	INT	-	NOT NULL	Foreign Key - EQUIPMENT	Equipment being maintained
StaffID	INT	-	NOT NULL	Foreign Key - STAFF	Staff performing maintenance
MaintenanceDate	DATE	-	NULL		Date of maintenance
Description	VARCHAR	200	NULL		Maintenance description

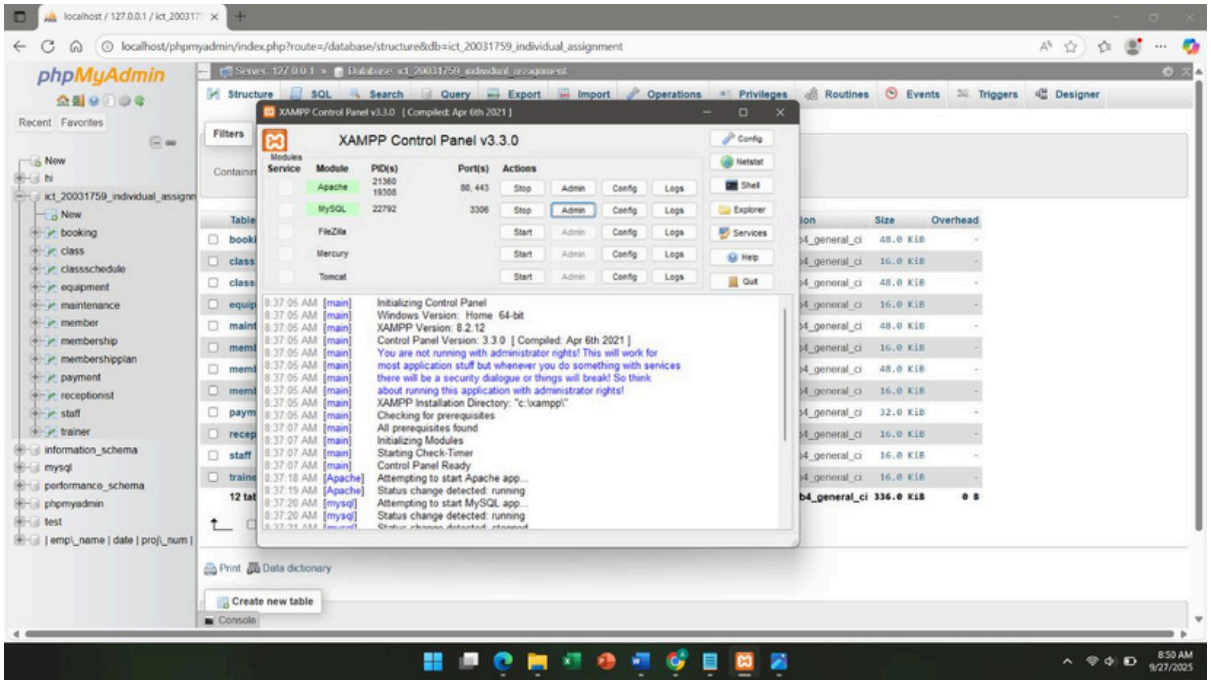
PAYMENT :

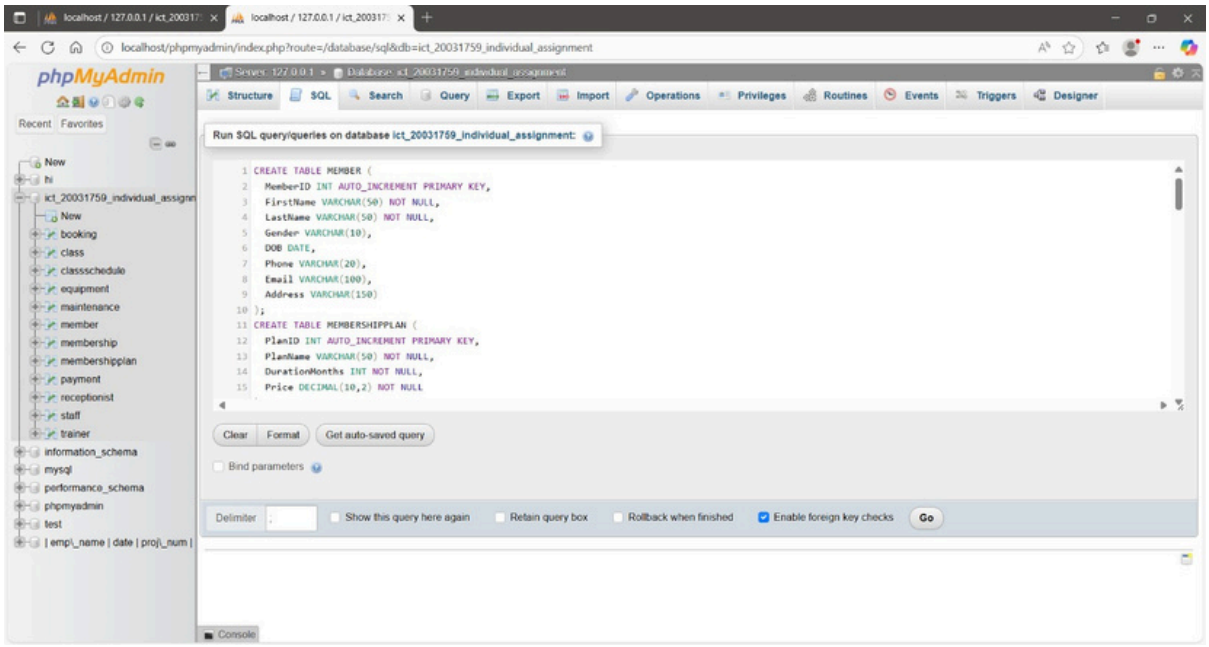
Attribute Name	Data Type	Size	Null / Not Null	Key / Constraint	Description
PaymentID	INT	-	NOT NULL	Primary Key (PK)	Unique payment ID
MembershipID	INT	-	NOT NULL	Foreign Key - MEMBERSHIP	Membership for which payment is made
PaymentDate	DATE	-	NULL		Date of payment

Amount	DECIMAL	10,2	NULL		Amount paid
Method	VARCHAR	50	NULL		Payment method

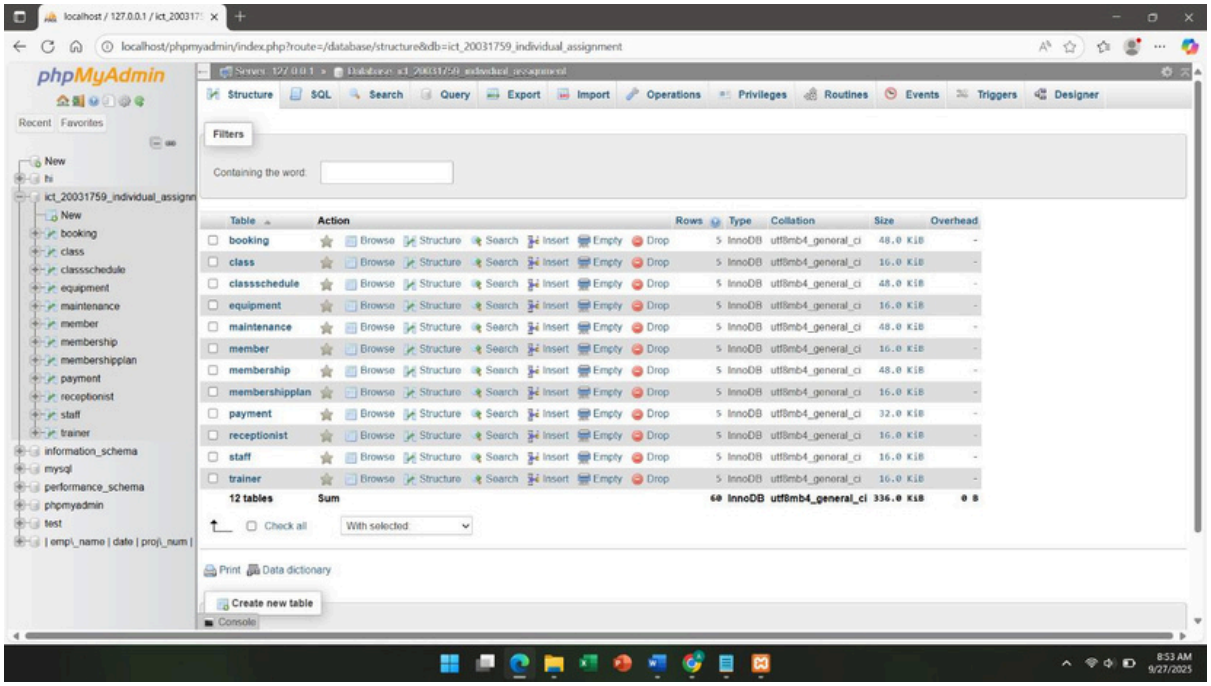
Screenshot of the XAMPP showing the creation of a database



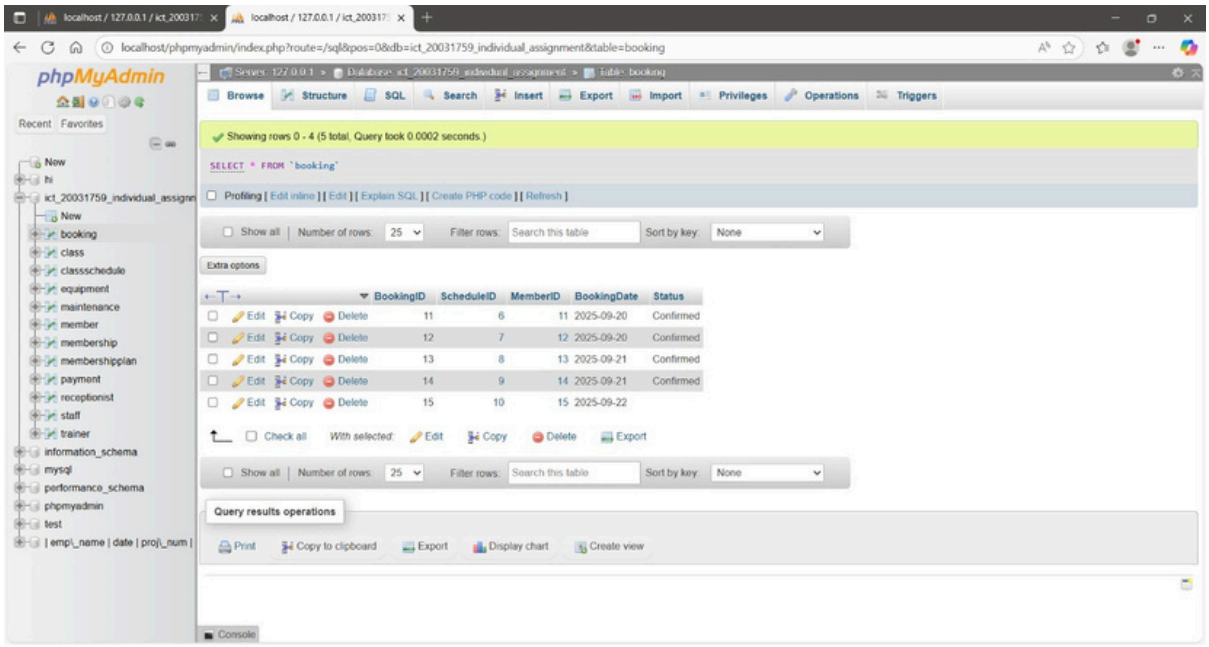

phpMyAdmin
XAMPP Control Panel v3.3.0
Apache
MySQL
FileZilla
Mercury
Tomcat

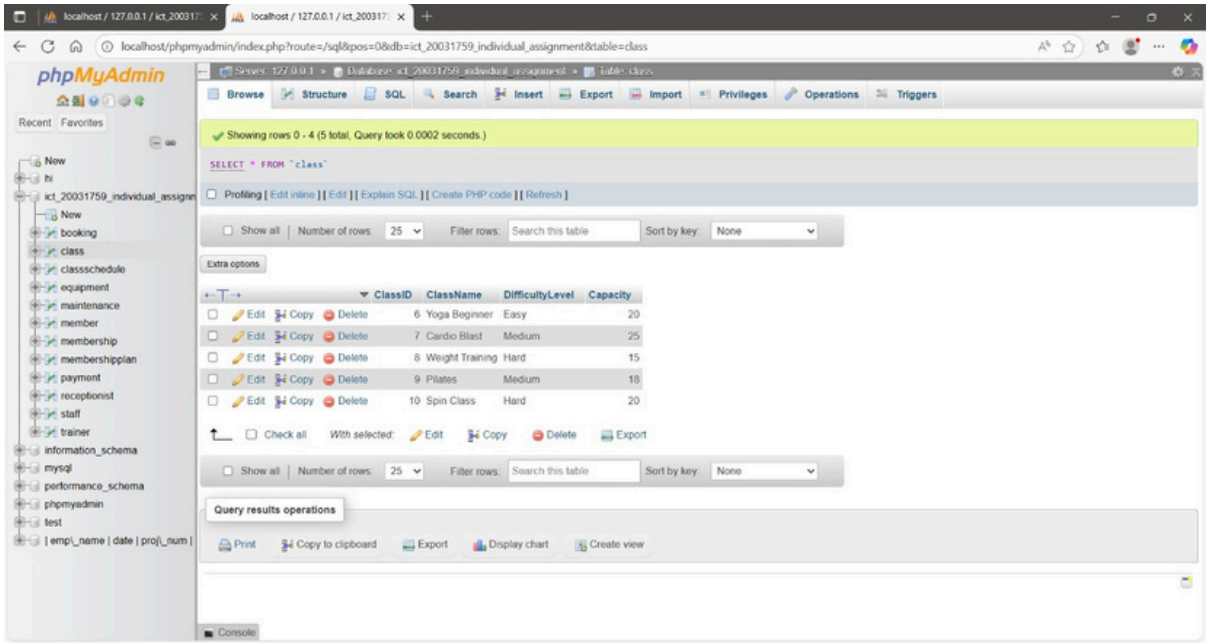

phpMyAdmin
Run SQL query/queries on database ict_20031759_individual_assignment:
CREATE TABLE MEMBER (
MemberID INT AUTO_INCREMENT PRIMARY KEY,
FirstName VARCHAR(50) NOT NULL,
LastName VARCHAR(50) NOT NULL,
Gender VARCHAR(10),
DOB DATE,
Phone VARCHAR(20),
Email VARCHAR(100),
Address VARCHAR(150)
);
CREATE TABLE MEMBERSHIPPLAN (
PlanID INT AUTO_INCREMENT PRIMARY KEY,
PlanName VARCHAR(50) NOT NULL,
DurationMonths INT NOT NULL,
Price DECIMAL(10,2) NOT NULL
Clear
Format
Get auto-saved query
Bind parameters
Delimiter
Show this query here again
Retain query box
Rollback when finished
Enable foreign key checks
Go

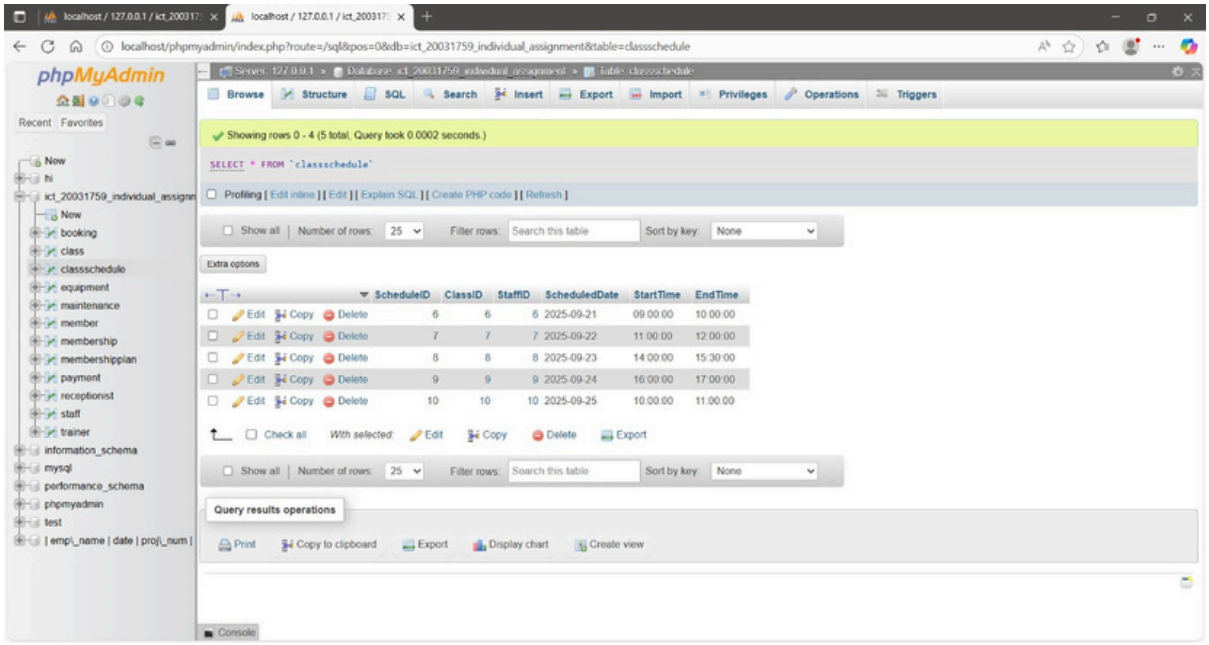
DDL includes screenshots from Xampp

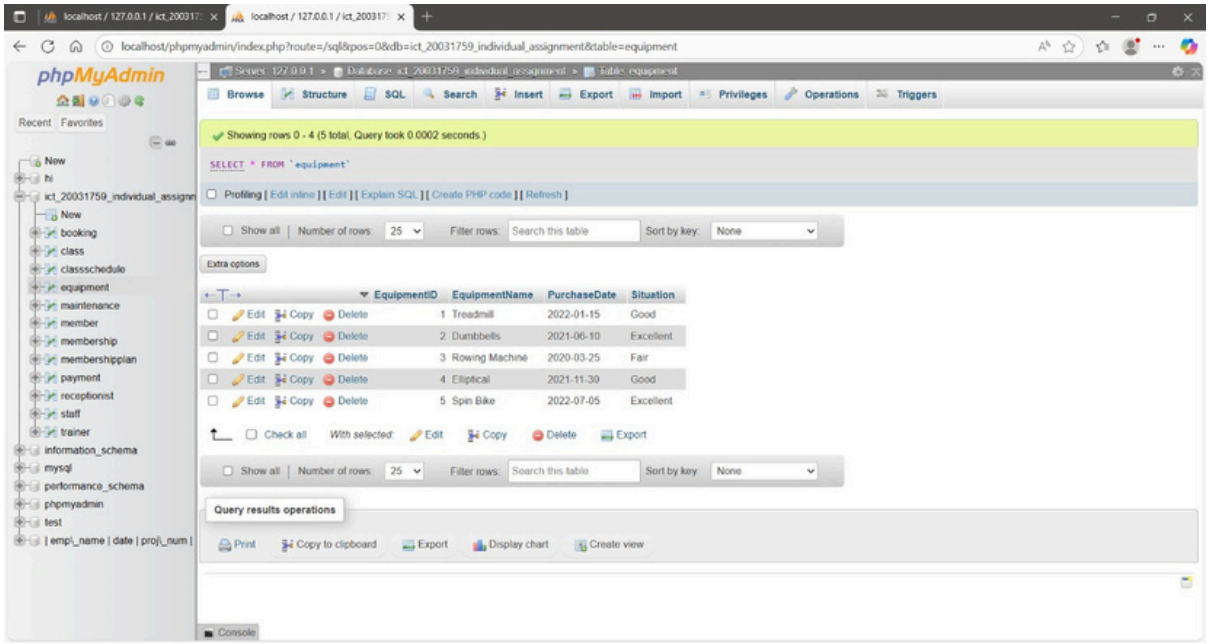


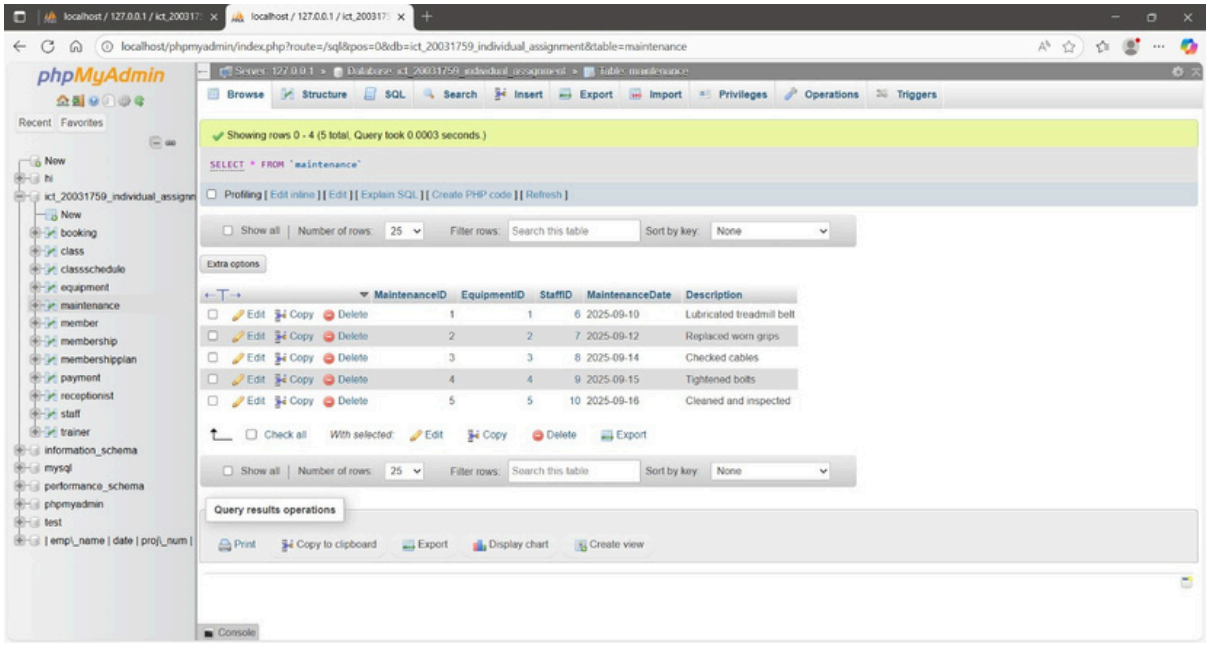
DML includes screenshots from Xampp

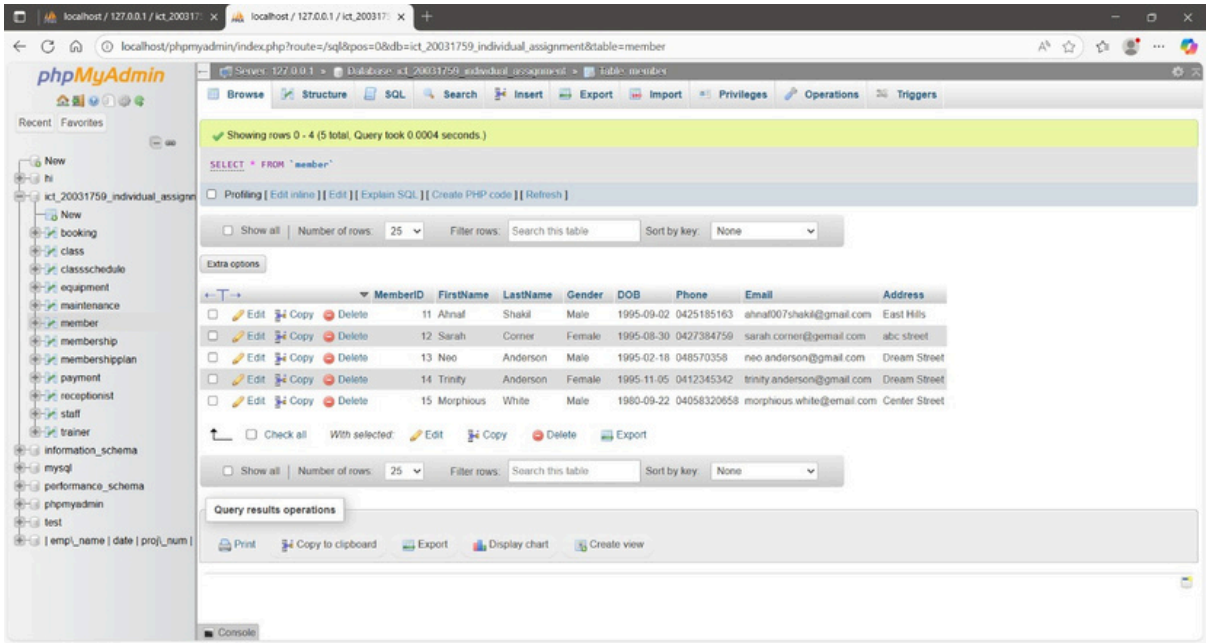


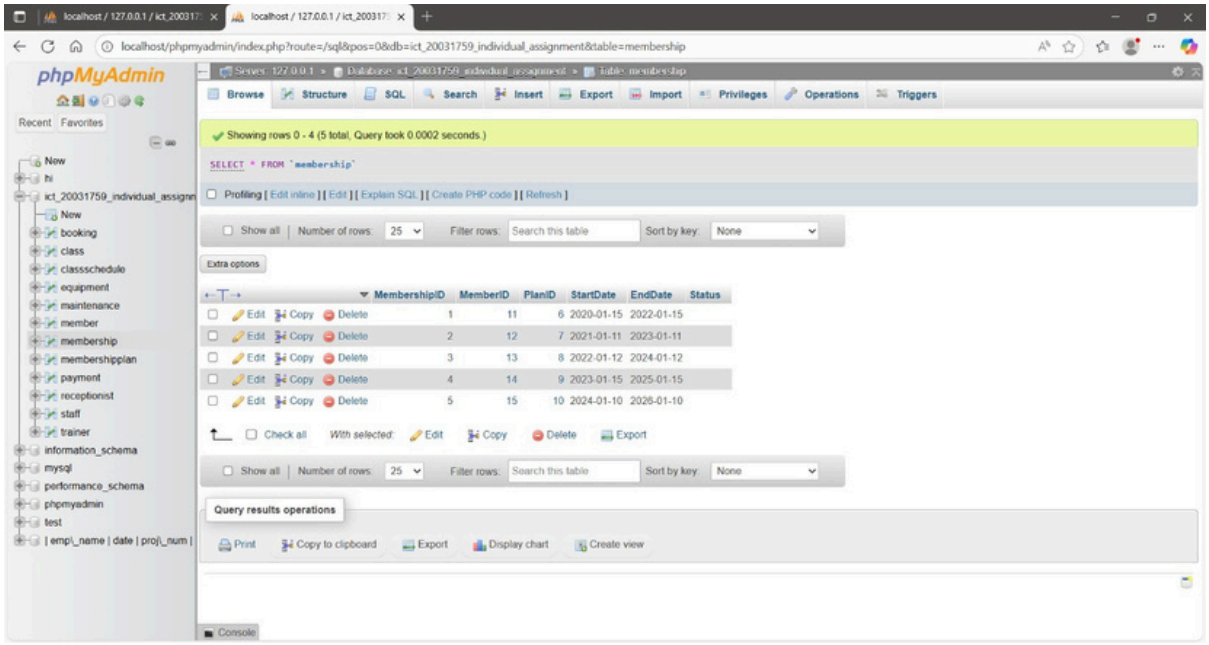

localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=class
Showing rows 0 - 4 (5 total, Query took 0.0002 seconds.)
SELECT * FROM `class`
ClassID ClassName DifficultyLevel Capacity
6 Yoga Beginner Easy 20
7 Cardio Blast Medium 25
8 Weight Training Hard 15
9 Pilates Medium 18
10 Spin Class Hard 20

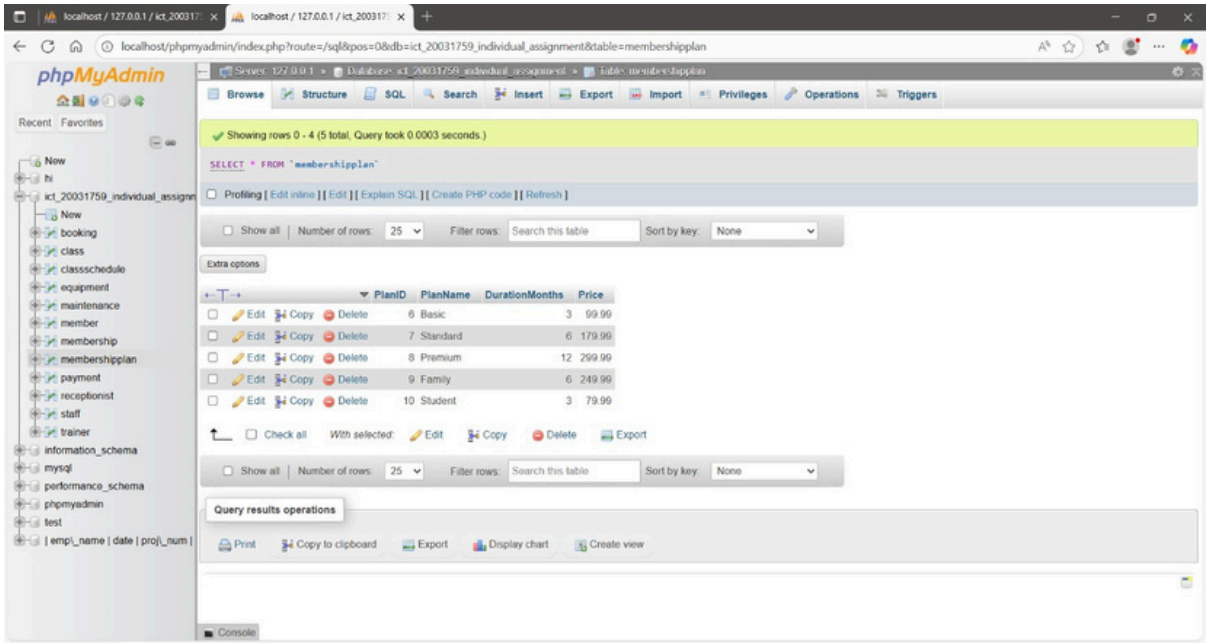

localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=classschedule
Showing rows 0 - 4 (5 total, Query took 0.0002 seconds.)
SELECT * FROM `classschedule`
ScheduleID ClassID StaffID ScheduledDate StartTime EndTime
6 6 6 2025-09-21 09:00:00 10:00:00
7 7 7 2025-09-22 11:00:00 12:00:00
8 8 8 2025-09-23 14:00:00 15:30:00
9 9 9 2025-09-24 16:00:00 17:00:00
10 10 10 2025-09-25 10:00:00 11:00:00

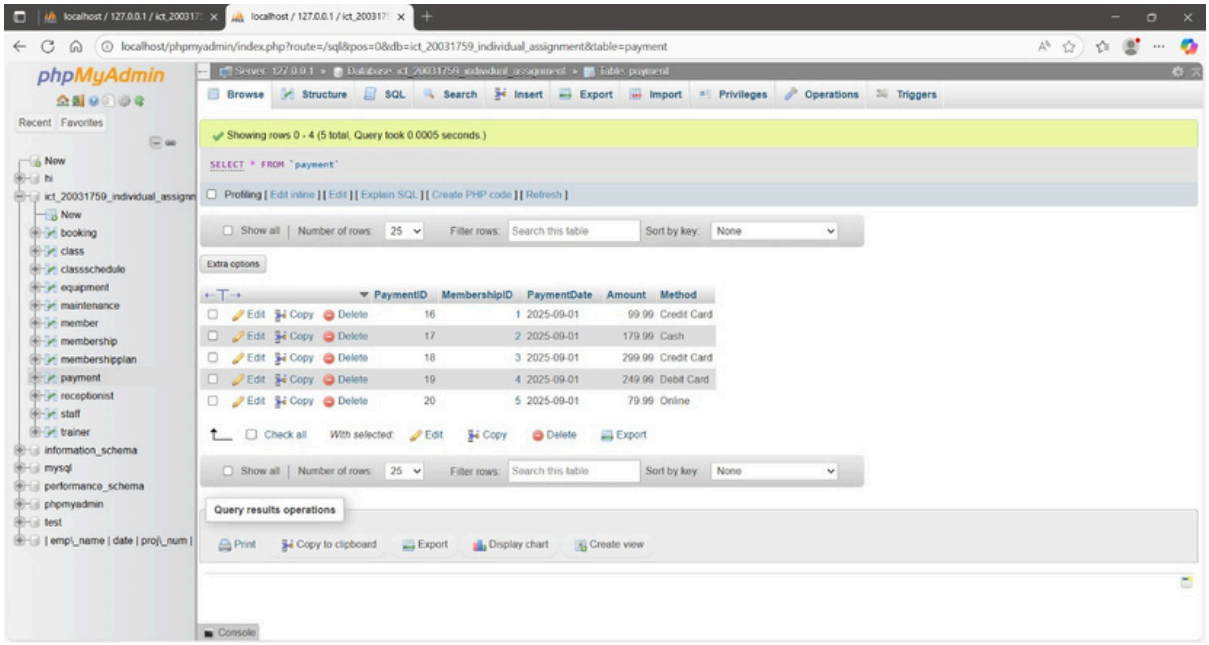

localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=equipment
phpMyAdmin
Browse Structure SQL Search Insert Export Import Privileges Operations Triggers
Showing rows 0 - 4 (5 total, Query took 0.0002 seconds.)
SELECT * FROM `equipment`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Extra options
EquipmentID EquipmentName PurchaseDate Situation
1 Treadmill 2022-01-15 Good
2 Dumbbells 2021-06-10 Excellent
3 Rowing Machine 2020-03-25 Fair
4 Elliptical 2021-11-30 Good
5 Spin Bike 2022-07-05 Excellent
Query results operations

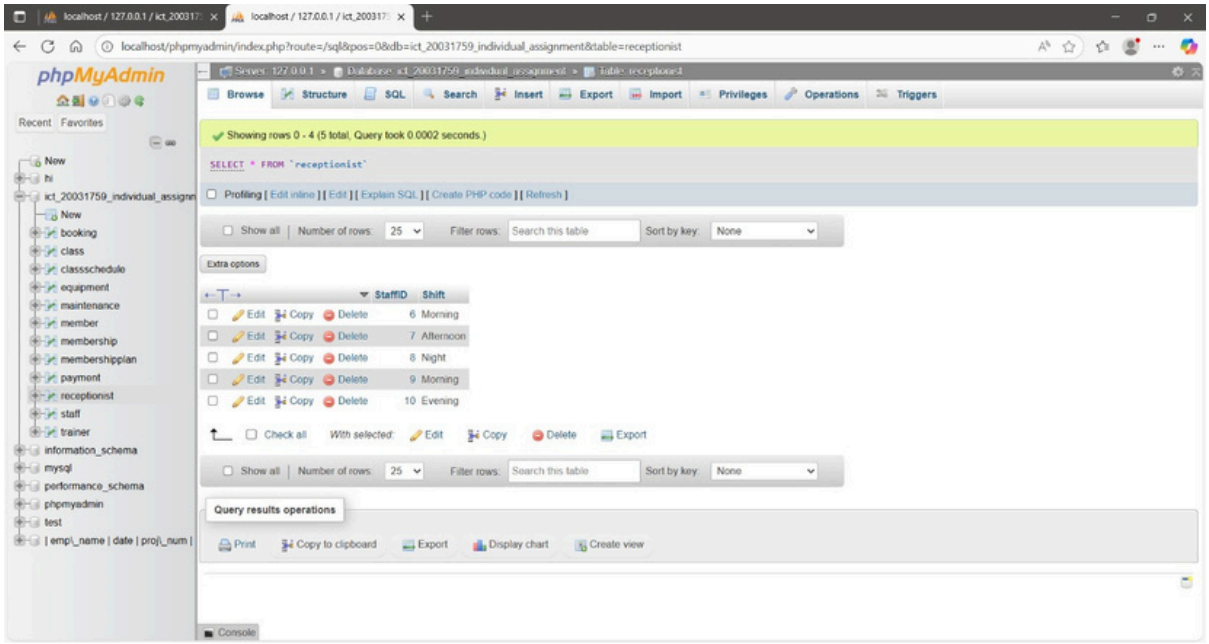

localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=maintenance
phpMyAdmin
Browse Structure SQL Search Insert Export Import Privileges Operations Triggers
Showing rows 0 - 4 (5 total, Query took 0.0003 seconds.)
SELECT * FROM `maintenance`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Extra options
MaintenanceID EquipmentID StaffID MaintenanceDate Description
1 1 6 2025-09-10 Lubricated treadmill belt
2 2 7 2025-09-12 Replaced worn grips
3 3 8 2025-09-14 Checked cables
4 4 9 2025-09-15 Tightened bolts
5 5 10 2025-09-16 Cleaned and inspected
Query results operations

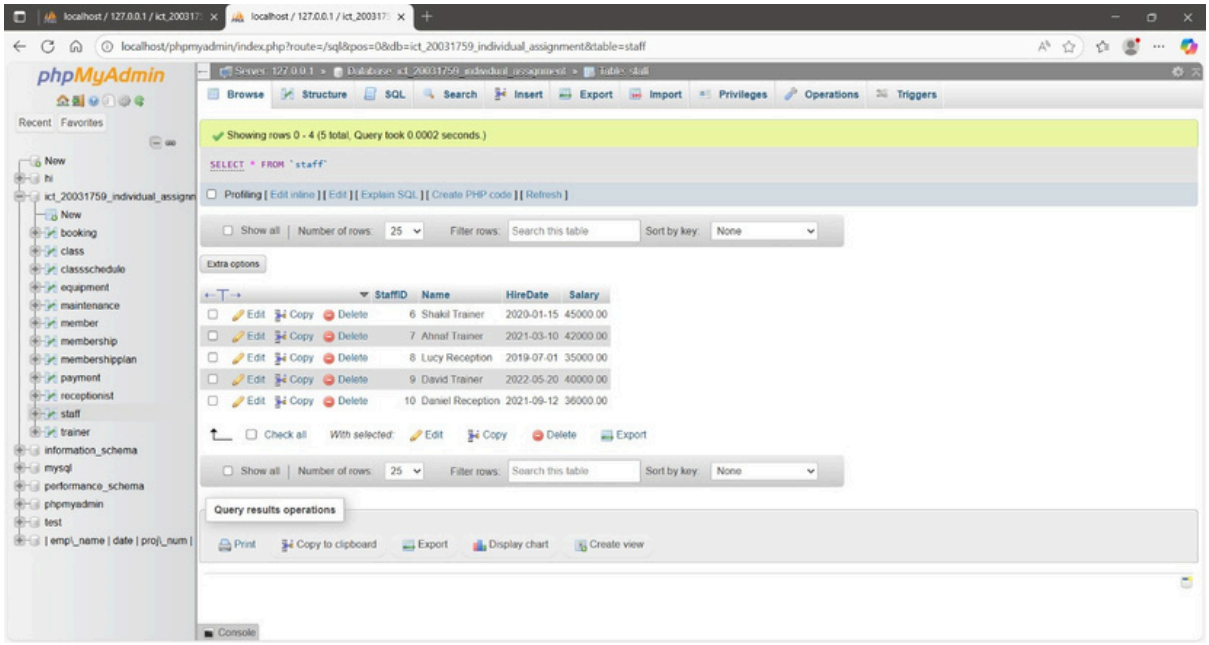

localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=member
phpMyAdmin
Browse Structure SQL Search Insert Export Import Privileges Operations Triggers
Showing rows 0 - 4 (5 total, Query took 0.0004 seconds.)
SELECT * FROM `member`
MemberID FirstName LastName Gender DOB Phone Email Address
11 Ahnaf Shakil Male 1995-09-02 0425185163 ahnaf007shakil@gmail.com East Hills
12 Sarah Corner Female 1995-08-30 0427384759 sarah.corner@gemail.com abc street
13 Neo Anderson Male 1995-02-18 048570358 neo.anderson@gmail.com Dream Street
14 Trinity Anderson Female 1995-11-05 0412345342 trinity.anderson@gmail.com Dream Street
15 Morphious White Male 1980-09-22 04058320658 morphious.white@email.com Center Street

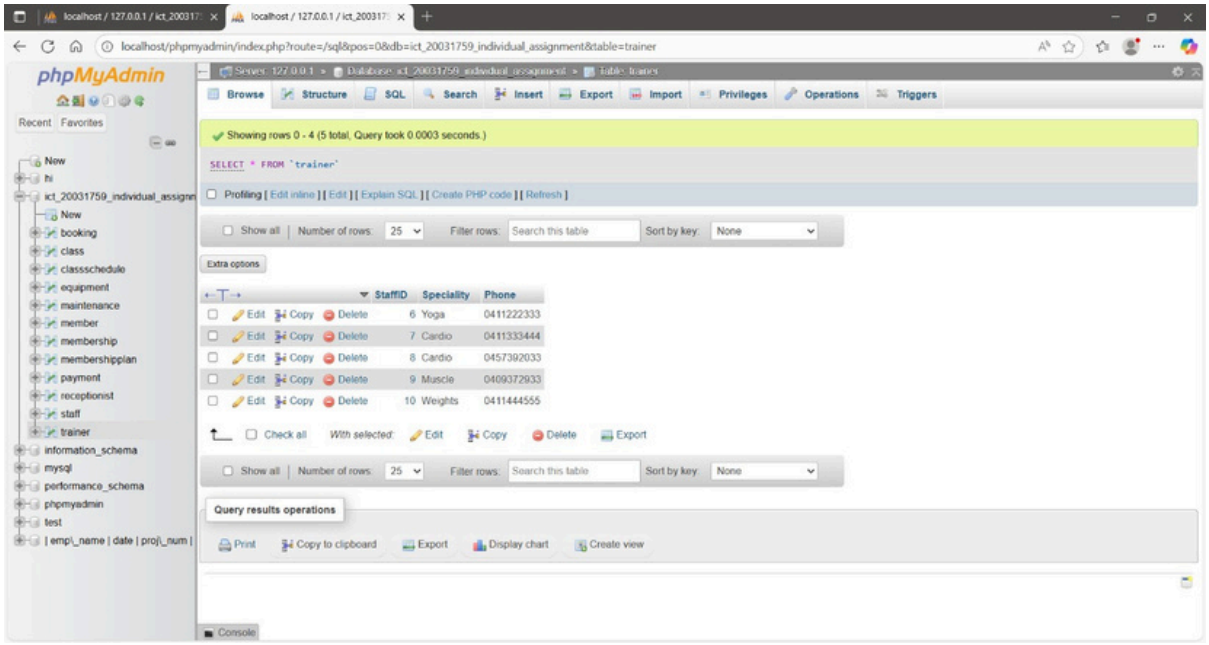

localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=membership
phpMyAdmin
Browse Structure SQL Search Insert Export Import Privileges Operations Triggers
Showing rows 0 - 4 (5 total, Query took 0.0002 seconds.)
SELECT * FROM `membership`
MembershipID MemberID PlanID StartDate EndDate Status
1 11 6 2020-01-15 2022-01-15
2 12 7 2021-01-11 2023-01-11
3 13 8 2022-01-12 2024-01-12
4 14 9 2023-01-15 2025-01-15
5 15 10 2024-01-10 2026-01-10


localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=membershipplan
Showing rows 0 - 4 (5 total, Query took 0.0003 seconds.)
SELECT * FROM `membershipplan`
PlanID PlanName DurationMonths Price
6 Basic 3 99.99
7 Standard 6 179.99
8 Premium 12 299.99
9 Family 6 249.99
10 Student 3 79.99


localhost/phpmyadmin/index.php?route=/sql&pos=0&db=ict_20031759_individual_assignment&table=payment
Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)
SELECT * FROM `payment`
PaymentID MembershipID PaymentDate Amount Method
16 1 2025-09-01 99.99 Credit Card
17 2 2025-09-01 179.99 Cash
18 3 2025-09-01 299.99 Credit Card
19 4 2025-09-01 249.99 Debit Card
20 5 2025-09-01 79.99 Online


localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=receptionist
phpMyAdmin
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Triggers
Showing rows 0 - 4 (5 total, Query took 0.0002 seconds.)
SELECT * FROM `receptionist`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
StaffID Shift
6 Morning
7 Afternoon
8 Night
9 Morning
10 Evening
Check all With selected: Edit Copy Delete Export
Query results operations
Print Copy to clipboard Export Display chart Create view
Console


localhost / 127.0.0.1 / ict_20031759_individual_assignment&table=staff
phpMyAdmin
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Triggers
Showing rows 0 - 4 (5 total, Query took 0.0002 seconds.)
SELECT * FROM `staff`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
StaffID Name HireDate Salary
6 Shakil Trainer 2020-01-15 45000.00
7 Ahnaf Trainer 2021-03-10 42000.00
8 Lucy Reception 2019-07-01 35000.00
9 David Trainer 2022-05-20 40000.00
10 Daniel Reception 2021-09-12 36000.00
Check all With selected: Edit Copy Delete Export
Query results operations
Print Copy to clipboard Export Display chart Create view
Console


localhost / 127.0.0.1 / ict_20031759
localhost/phpmyadmin/index.php?route=/sql&pos=0&db=ict_20031759_individual_assignment&table=trainer
phpMyAdmin
Recent Favorites
New
hi
ict_20031759_individual_assignm
New
booking
class
classschedule
equipment
maintenance
member
membership
membershipplan
payment
receptionist
staff
trainer
information_schema
mysql
performance_schema
phpmyadmin
test
| emp_name | date | proj_num |
Server: 127.0.0.1 » Database: ict_20031759_individual_assignment » Table: trainer
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Triggers
Showing rows 0 - 4 (5 total, Query took 0.0003 seconds.)
SELECT * FROM `trainer`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
StaffID Speciality Phone
Edit Copy Delete 6 Yoga 0411222333
Edit Copy Delete 7 Cardio 0411333444
Edit Copy Delete 8 Cardio 0457392033
Edit Copy Delete 9 Muscle 0409372933
Edit Copy Delete 10 Weights 0411444555
Check all With selected: Edit Copy Delete Export
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Query results operations
Print Copy to clipboard Export Display chart Create view
Console

The end